

TinyF和RDK X3串口通信

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0.TinyF简介

- TinyF 激光测距模组是基于 D-TOF 单点激光传感器，板载 mcu，并进行外壳封装而开发的测距模组，内置阳光抑制和抗盖板脏污算法，能精准快速地测量目标物距离。在 100-190K Lux 的超强环境光中，能做到精准测量极小目标物，且不受目标物反射率差异而影响 2m~4m精准测距效果。

1.实验准备

- TinyF激光测距模组
- RDK系列 本教程实验的是RDK X3

2.实验接线

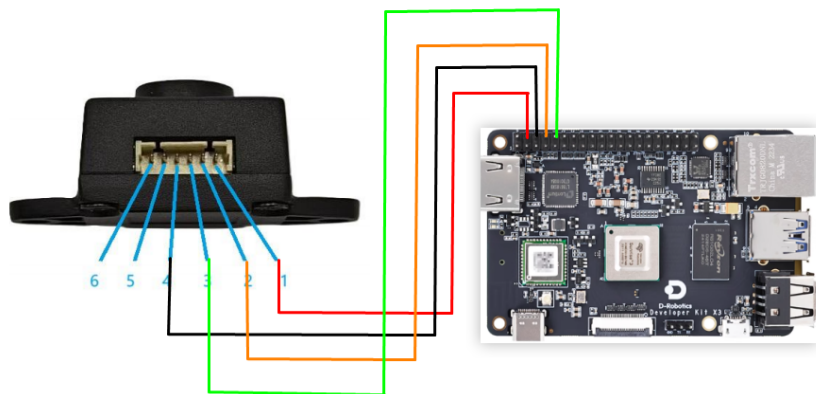
RDK X3	TinyF
UART_TXD	RX
UART_RXD	TX
GND	GND
5V	5V

接线如图

RDK X3 40Pin Pinout Diagram									
Function Description	X3 Pin Number	BCM Code	CVM Function Name	Physical Pin BOARD Code		CVM Function Name	BCM Code	X3 Pin Number	Function Description
3.3V Power Signal			VDD_3V3	1	2	VDD_5V			5V Power Signal
I2C0 Data Signal	9	2	I2C0_SDA	3	4	VDD_5V			5V Power Signal
I2C0 Clock Signal	8	3	I2C0_SCL	5	6	GND			GND Signal
I2S0 MCLK Clock Signal	101	4	I2S0_MCLK	7	8	UART_TXD	14	111	UART3 Transmit Signal
GND Signal			GND	9	10	UART_RXD	15	112	UART3 Receive Signal
GPIO17 Signal	12	17	GPIO17	11	12	I2S0_BCLK	18	102	I2S0 BCLK Clock Signal
GPIO27 Signal	13	27	GPIO27	13	14	GND			GND Signal
GPIO22 Signal	30	22	GPIO22	15	16	GPIO23	23	27	GPIO23 Signal
3.3V Power Signal			VDD_3V3	17	18	GPIO24	24	22	GPIO24 Signal
SPI1 MOSI Signal	6	10	SPI1_MOSI	19	20	GND			GND Signal
SPI1 MISO Signal	7	9	SPI1_MISO	21	22	GPIO25	25	29	GPIO25 Signal
SPI1 CLK Signal	3	11	SPI1_SCLK	23	24	SPI1_CSN	8	5	SPI1 CS Signal
GND Signal			GND	25	26	GPIO7	7	28	GPIO7 Signal
I2C3 Data Signal	15	0	I2C3_SDA	27	28	I2C3_SCL	1	14	I2C3 Clock Signal
GPIO5 Signal	119	5	GPIO5	29	30	GND			GND Signal
GPIO6 Signal	118	6	GPIO6	31	32	PWM4	12	25	PWM4 Signal
PWM0 Signal	4	13	PWM0	33	34	GND			GND Signal
I2S0 LRCK Signal	103	19	I2S0_LRCK	35	36	GPIO16	16	20	GPIO16 Signal
GPIO26 Signal	117	26	GPIO26	37	38	I2S1_SDIO	20	108	I2S1_SDIO Signal
GND Signal			GND	39	40	I2S0_SDIO	21	104	I2S0_SDIO Signal

RDK X3	TinyF
UART_TXD	RX
UART_RXD	TX
GND	GND
5V	5V

1	VDD 5V
2	RX
3	TX
4	GND
5	NA
6	NA



3.主要代码讲解

```
def parse_data_format(self, data_bytes):
    try:
        # 查找逗号分隔符的位置 (0x2C) Find the position of the comma separator (0x2C)
        comma_index = -1
        for i, b in enumerate(data_bytes):
            if b == 0x2c:
                comma_index = i
                break

        if comma_index == -1:
            return None, None

        # 提取距离部分 (逗号前的数据) Extract the distance part (the data before the comma)
        # 跳过开头的空格(0x20) skip the initial space (0x20)
        start_index = 0
        while start_index < comma_index and data_bytes[start_index] == 0x20:
            start_index += 1
```

```

# 距离数据长度 (2-5字节) Distance data length (2-5 bytes) :
distance_bytes = data_bytes[start_index:comma_index]

# 提取置信度部分 (逗号后的数据) Extract the confidence part (the data
after the comma)
# 跳过逗号后的空格(0x20) The space after the comma (0x20)
confidence_start = comma_index + 1
while confidence_start < len(data_bytes) and
data_bytes[confidence_start] == 0x20:
    confidence_start += 1

# 置信度数据应为1-2字节:原始数据 The confidence data should be 1-2 bytes:
raw data
confidence_bytes = data_bytes[confidence_start:len(data_bytes)-1]

# 转换距离数据 Convert distance data
distance_str = distance_bytes.decode('ascii')
distance = int(distance_str)
if distance >4000:
    return 0,0

# 转换置信度数据 Convert the confidence level data
confidence_str = confidence_bytes.decode('ascii')
confidence = int(confidence_str)

return distance, confidence
except Exception as e:
    print(f"parse_error: {e}")
    return None, None

....

if __name__ == "__main__":
    # 根据设备实际串口号进行替换 Replace according to the actual serial port number
of the device
    SERIAL_PORT = "/dev/ttyS3"
    BAUD_RATE = 115200
    parser = DistanceDataParser(serial_port=SERIAL_PORT, baud_rate=BAUD_RATE)
    parser.start_processing()

```

以上程序为解析TinyF 激光测距模组数据的功能，只有符合协议才可以正确解析出数据。

4.实验现象

- 1.在断电的情况下，按照接线图接好线。
- 2.如果Python3里没有安装serial串口驱动库，请运行以下命令安装。

```
pip3 install pyserial
```

- 3.需要提前把uart_test.py程序上传到RDK X3上，然后终端输入以下命令

```
python3 uart_test.py
```

- 4.运行命令后，终端会打印测距信息

```
Terminal - root@ubuntu: ~
File Edit View Terminal Tabs Help
Distance: 434 mm, Confidence: 58
Distance: 437 mm, Confidence: 58
Distance: 438 mm, Confidence: 58
Distance: 781 mm, Confidence: 56
Distance: 450 mm, Confidence: 54
Distance: 781 mm, Confidence: 57
Distance: 782 mm, Confidence: 57
Distance: 452 mm, Confidence: 58
Distance: 445 mm, Confidence: 58
Distance: 434 mm, Confidence: 58
Distance: 430 mm, Confidence: 58
Distance: 430 mm, Confidence: 58
Distance: 434 mm, Confidence: 50
Distance: 707 mm, Confidence: 58
Distance: 689 mm, Confidence: 58
Distance: 670 mm, Confidence: 58
Distance: 601 mm, Confidence: 59
Distance: 585 mm, Confidence: 58
Distance: 358 mm, Confidence: 51
Distance: 267 mm, Confidence: 58
```

5.TinyF一些重要参数

相关参数说明

名称	值
波特率	115200
距离有效最小值（理论值）	20, 单位 mm
距离有效最大值（理论值）	4000, 单位 mm
置信度	0~62, 值越大表示距离可信度越高

数据格式

帧结构: [0x20][distance][0x2C][0x20][confidence][0x0A]

示例: 20 33 32 37 2C 20 36 31 0A

字段	字节数	示例 (Hex)	ASCII 字符
帧头 (固定值)	1	20	空格
距离值 (distance)	1~5	33 32 37	327
分隔符 (固定值)	2	2C 20	,空格
置信度 (confidence)	1~2	36 31	61
帧尾 (固定值)	1	0A	\n

光照度	测试卡	测距范围	精度	准度
室内 0Klux	白板 88%	2cm-5.0m	<u>< 40mm@ < 3m</u> <u>< 20mm@ > 3m</u>	< 3m±2% > 3m±1%
室内 0Klux	灰板 18%	2cm-5.0m	<u>< 40mm@ < 3m</u> <u>< 20mm@ > 3m</u>	< 3m±2% > 3m±1%
室外 100Klux	白板 88%	2cm-4.4m	< 70mm@ < 3m < 60mm@ > 3m	< 3m±2.5% > 3m±2%
室外 100Klux	灰板 18%	2cm-3.9m	< 90mm@ < 3m < 50mm@ > 3m	< 3m±3% > 3m±2%
室外 190Klux	白板 88%	2cm-2.9m	< 60mm@ < 1m	±6%
室外 190Klux	灰板 18%	2cm-0.9m	< 80mm@ < 1m	±4%